

Exploratory Data Analysis

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CST-411

Open dataset used

Student Depression Dataset

- Aimed at understanding, analyzing, and predicting depression levels among students. Provides insights into factors that contribute to student mental health challenges.

What we looked for

- Whether any of the participants had suicidal thoughts.
- The correlation and difference between financial stress and depression.
- Correlation between other factors (job satisfaction, study satisfaction, etc)

Import data

```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

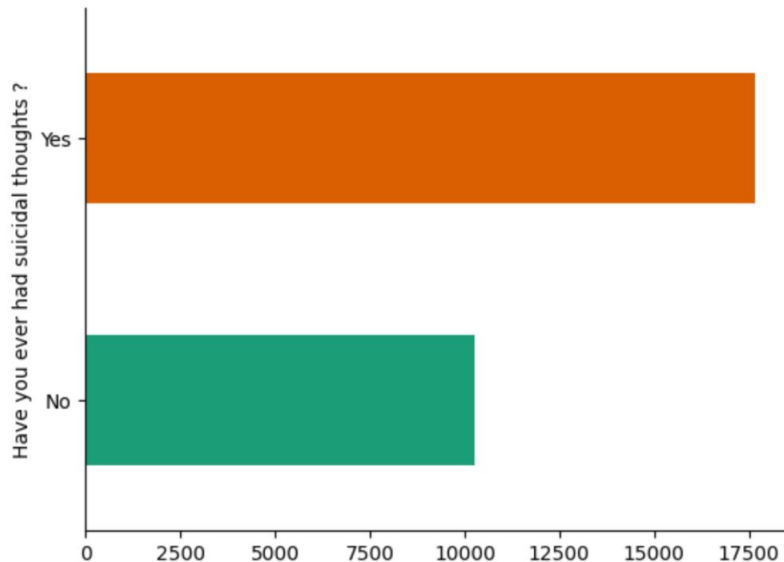
```
[ ] df = pd.read_csv('student_depression_dataset.csv')
df.head()
df.tail()
```

City	Profession	Academic Pressure	Work Pressure	CGPA	Study Satisfaction	Job Satisfaction	Sleep Duration	Dietary Habits	Degree	Have you ever had suicidal thoughts ?	Work/Study Hours	Financial Stress	Family History of Mental Illness	Depression
Visakhapatnam	Student	5.0	0.0	8.97	2.0	0.0	'5-6 hours'	Healthy	B.Pharm	Yes	3.0	1.0	No	1
Bangalore	Student	2.0	0.0	5.90	5.0	0.0	'5-6 hours'	Moderate	BSc	No	3.0	2.0	Yes	0
Srinagar	Student	3.0	0.0	7.03	5.0	0.0	'Less than 5 hours'	Healthy	BA	No	9.0	1.0	Yes	0
Varanasi	Student	3.0	0.0	5.59	2.0	0.0	'7-8 hours'	Moderate	BCA	Yes	4.0	5.0	Yes	1
Jaipur	Student	4.0	0.0	8.13	3.0	0.0	'5-6 hours'	Moderate	M.Tech	Yes	1.0	1.0	No	0

Results

▼ Have you ever had suicidal thoughts ?

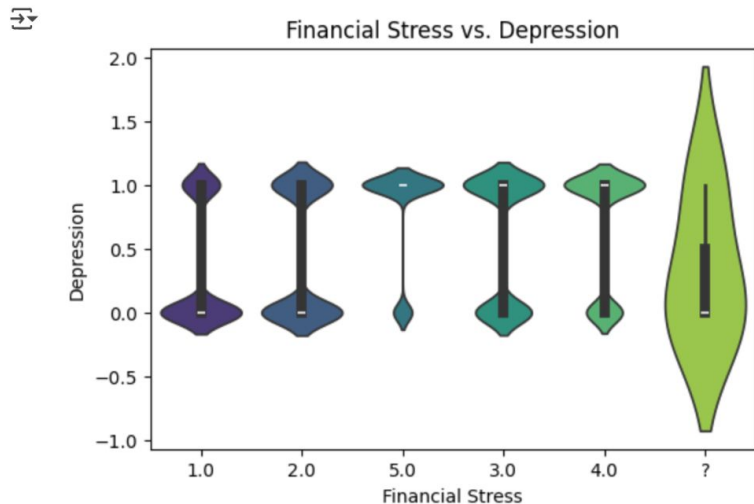
```
# @title Have you ever had suicidal thoughts ?  
  
from matplotlib import pyplot as plt  
import seaborn as sns  
df.groupby('Have you ever had suicidal thoughts ?').size().plot(kind='barh', color=sns.palettes.mpl_palette('Dark2'))  
plt.gca().spines[['top', 'right',]].set_visible(False)
```



```
[ ] df.columns
```

```
Index(['id', 'Gender', 'Age', 'City', 'Profession', 'Academic Pressure',  
      'Work Pressure', 'CGPA', 'Study Satisfaction', 'Job Satisfaction',  
      'Sleep Duration', 'Dietary Habits', 'Degree',  
      'Have you ever had suicidal thoughts ?', 'Work/Study Hours',  
      'Financial Stress', 'Family History of Mental Illness', 'Depression'],  
      dtype='object')
```

```
[ ] plt.figure(figsize=(6, 4))  
sns.violinplot(x='Financial Stress', y='Depression', data=df, palette='viridis')  
plt.title("Financial Stress vs. Depression")  
plt.show()
```



```

numeric_df = df.select_dtypes(include=['number'])
plt.figure(figsize=(8, 6))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title("Correlation Heatmap")
plt.show()

```

